Book 1 — Lecture 1 — Exercise 3

Allowable Laws on the Integer Line

Prompt. Determine which of the dynamical laws shown in Eqs. (2) through (5) are *allowable*. Equation (1), $N(n+1) = N(n) - 1$, is given as an allowable example.

What "allowable" means

A law is allowable when it is:

1. **Deterministic**: each current state has one successor.[1]
2. **Reversible**: each state has one predecessor[2] (bijection[3] on $\mathbb{Z}$[4]).

Equations

$$
\begin{aligned}
&(1) \quad N(n+1) = N(n) - 1, \\
&(2) \quad N(n+1) = N(n) + 1, \\
&(3) \quad N(n+1) = N(n) + 2, \\
&(4) \quad N(n+1) = N(n)^2, \\
&(5) \quad N(n+1) = (-1)^{N(n)} N(n).
\end{aligned}
$$

Verdicts

(2) $N(n+1) = N(n) + 1$: allowable.
It is the inverse-time version of Eq. (1), with inverse $N \mapsto N - 1$.

(3) $N(n+1) = N(n) + 2$: allowable.
It is a bijection with inverse $N \mapsto N - 2$.

(4) $N(n+1) = N(n)^2$: not allowable.
Not injective[5] ($2^2 = (-2)^2$) and not surjective[6] onto $\mathbb{Z}$.

[1] A *successor* is the state reached in one forward update.
[2] A *predecessor* is a state that maps to the current one under the update rule.
[3] A *bijection* is both one-to-one and onto: every target has exactly one source.
[4] $\mathbb{Z}$ denotes all integers: $\ldots, -2, -1, 0, 1, 2, \ldots$.
[5] *Injective* means different inputs give different outputs.
[6] *Surjective* means every target value is hit by at least one input.

(5) $N(n+1) = (-1)^{N(n)} N(n)$**: allowable.**
Even integers are fixed; odd integers map to their negative, giving 2-cycles. Each state has a unique predecessor.

Summary

Equation	Allowable?
(2) $N(n+1) = N(n) + 1$	Yes
(3) $N(n+1) = N(n) + 2$	Yes
(4) $N(n+1) = N(n)^2$	No
(5) $N(n+1) = (-1)^{N(n)} N(n)$	Yes

Result. Allowable: (2), (3), and (5). Not allowable: (4).

Example. Eq. (4) fails reversibility immediately because $2^2 = (-2)^2 = 4$, so the predecessor of 4 is not unique.